Lab 6

Objective: The purpose of this lab is to practice working with **std::string** and **std::vector** in C++. By the end of this lab, you should be able to perform operations like modifying strings, searching within strings, manipulating vectors, and using common methods associated with these types.

Part 1: String Manipulation

Use the following reference to finish the task highlighted in red
https://cplusplus.com/reference/string/string/

```
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;
int main(){

  string str="Problem Solving with C++";
  transform(str.begin(), str.end(), str.begin(), ::toupper);//transform str to uppercase
  cout<<str<<endl;

  int pos; //variable for position

  // find the position of “LEM”

  if (pos != string::npos) {

      cout << "Position of 'LEM': " << pos << endl;
   }

   pos = str.find("WITH");
   if (pos != string::npos) {

   // Replace "WITH" with "WITH AMAZING"

   }
   cout << "Modified Sentence: " << str << endl;
  return 0;
}
```

Part 2: Vector Manipulation

Task 2.1: Working with Integer Vectors

Use the following reference to finish the task highlighted in red
https://cplusplus.com/reference/vector/vector/

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
```

```
int main() {
    vector<int> numbers = {10, 20, 30, 40, 50};

    // Add 60 to the end of the vector, use push_back()


    // Insert 25 between 20 and 30, use insert()



    // Sort in descending order
    sort(numbers.begin(), numbers.end());

    // Print the result
    cout << "Sorted Vector: ";
    for (int num : numbers) {
        cout << num << " ";
    }
    cout << endl;

    return 0;
}
```

Task 2.2: String Vector Operations

- Create a vector of strings called words and initialize it with values such as {"apple", "banana", "cherry", "date"}.
- Add "elderberry" to the vector.
- Sort the vector in alphabetical order.
- Search for "cherry" and print its position in the vector.
- Remove "banana" from the vector and print the final list of words.

Submission:

You need to upload to Moodle: the source code (.cpp files), and the screenshot of the running of your program before 4pm Wednesday.